Chapter 4 — Work, Energy and Power

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Q.1 — Multiple Choice Questions (with correct option and reason)

4.1. A 1 kg mass has potential energy of 1 J relative to ground at height:

(a) 0.102 m **(b)** 1 m
(c) 9.8 m **(d)** 32 m

Answer: (a) 0.102 m — h = U/mg = 1/9.8 ≈ 0.102 m.

4.2. Iron and aluminium spheres of same diameter dropped from a cliff have identical, at 10 m from ground:

(a) accelerations **(b)** momentums
(c) potential energies **(d)** kinetic energies

Answer: (a) accelerations — Acceleration due to gravity is the same; masses differ.

4.3. A body at rest may have:

(a) speed **(b)** velocity
(c) momentum **(d)** energy

Answer: (d) energy — It can have potential or internal energy.

4.4. A child on a swing varies from 0.5 m to 1.5 m. Maximum speed is about:

(a) 1.5 m s^{-1} **(b)** 4.4 m s^{-1}
(c) 9.8 m s^{-1} **(d)** depends on mass

Answer: (b) 4.4 m s^{-1} — $v = \sqrt{(2g\Delta h)} = \sqrt{(19.6)} \approx 4.4$ m/s.

4.5. When a ball is thrown up and falls back, the conservative force is:

(a) air resistance **(b)** gravity
(c) friction between ball and air **(d)** contact force with hand

Answer: (b) gravity — Gravity is conservative.

4.6. According to work-energy principle, work done on a body equals:

(a) change in K.E. **(b)** change in P.E.
(c) zero **(d)** sum of K.E. and P.E.

Answer: (a) change in K.E. — $W_{net} = \Delta K$.

4.7. A 6 W lamp in 2 min gives out energy:

(a) 12 J **(b)** 20 J
(c) 360 J **(d)** 720 J

Answer: (d) 720 J — E = Pt = 6 × 120 = 720 J.

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4.8. A battery delivers 3000 J to a 2 W motor. It runs for:

(a) 1500 min **(b)** 10 min
(c) 50 min **(d)** 25 min

Answer: (d) 25 min — t = 3000/2 = 1500 s = 25 min.

4.9. K.E. acquired from rest over fixed distance under constant force is:

(a) proportional to √m **(b)** proportional to m
(c) zero **(d)** independent of m

Answer: (d) independent of m — W = Fd = K, independent of mass.

4.10. A body moves 10 m under 5 N force; work done is 25 J. The angle is:

(a) 0° **(b)** 30°
(c) 60° **(d)** 90°

Answer: (c) 60° — cos θ = 25/(5×10) = 0.5.

Q.2 — Short Answer Questions

4.1. Why is electrical power required when an elevator descends? Why limit passengers?

Power is needed for control, braking and safety systems, and to manage the rate of descent. Limiting passengers limits the gravitational energy that must be controlled or dissipated, keeping the system safe.

4.2. A body is raised to height H. What is the sign of work done by the body and Earth?

The external agent does positive work on the body, while gravity does negative work. The body–Earth system gains gravitational potential energy.

4.3. A body falls towards Earth in air. Is total mechanical energy conserved?

No. Air resistance converts some mechanical energy into heat and sound, so mechanical energy decreases, although total energy is conserved.

4.4. Calculate power of a crane lifting 1000 kg to 100 m in 20 s.

P = mgh/t = 1000 × 9.8 × 100 / 20 = 49,000 W = 49 kW.

4.5. A trolley with sand bags moves on a frictionless track; sand leaks out. What happens to speed?

The speed remains unchanged. The leaking sand carries away the same horizontal velocity, and no external horizontal force acts, so the trolley's speed does not change.

4.6. Give absolute and gravitational units of work in MKS and CGS.

MKS: absolute unit joule, gravitational unit kgf·m. CGS: absolute unit erg, gravitational unit gf·cm.

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4.7. A body dropped from H reaches ground with speed 1.2√(gH). Find work done by air friction.

K = ½m(1.44gH) = 0.72mgH. Loss of potential energy is mgH, so work done by friction = 0.72mgH − mgH = −0.28mgH.

4.8. A bicycle has K.E. 150 J. Find K.E. if (i) same mass, double speed (ii) triple mass, half speed.

(i) $K \propto v^2$, so $150 \times 4 = 600$ J. (ii) $K \propto mv^2$, so $150 \times 3 \times ¼ = 112.5$ J.

4.9. What happens to K.E. when momentum becomes double?

$K = p^2/2m$, so doubling momentum makes kinetic energy four times.

4.10. Does the international space station have gravitational P.E., K.E. or both?

Both. It moves with orbital speed, so it has kinetic energy, and it is in Earth's gravitational field, so it has gravitational potential energy.

Constructed Response Questions

4.1 When is a force conservative? Give two conditions.

A force is conservative if (1) work done depends only on initial and final positions, not path, and (2) work done over any closed path is zero.

4.2 A light and a heavy body have same linear momentum. Which has greater K.E.?

$K = p^2/2m$. For the same momentum, the lighter body has greater kinetic energy.

4.3 A motorcycle runs at constant speed on a horizontal track with no net force. Is any work being done?

No net work is done on the motorcycle because its kinetic energy does not change. The engine may do work against friction, but net work is zero.

4.4 A force reduces a ball's speed from 14 to 6 m s^{-1}. Is work positive or negative?

Negative work, because kinetic energy decreases.

4.5 How can a slow truck have more K.E. than a fast car?

Kinetic energy depends on mass as well as speed. A very large mass can give a slow truck more K.E. than a lighter fast car.

4.6 Why is work done against friction non-conservative?

It depends on the path length and is not recovered when the path is reversed; it is converted into heat, so work over a closed path is not zero.

4.7 Does wind contain kinetic energy?

Yes. Wind is moving air, and moving mass has kinetic energy $½mv^2$.

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Comprehensive / Long Questions

4.1 Define K.E. and derive an expression for it.

Kinetic energy is the energy of motion. A force F accelerates a mass m from rest through distance d, reaching speed v. Using $v^2 = 2ad$, $a = v^2/2d$. Work $W = Fd = mad = \frac{1}{2}mv^2$, so $K = \frac{1}{2}mv^2$.

4.2 How is work done by (i) a constant force (ii) a variable force?

(i) $W = Fd\cos\theta$. (ii) Divide the path into small strips; work is the sum of $F\cos\theta\,\Delta d$, equal to the area under the $F\cos\theta$–d graph.

4.3 Define conservative field. Show gravitational field is conservative.

A conservative field is one in which work done is path independent and zero over a closed path. For gravity near Earth, work depends only on height change: $W = -mg\Delta h$. Over a closed path $\Delta h = 0$, so work is zero.

4.4 What is absolute P.E.? Derive an expression for it.

Absolute gravitational potential energy is measured with zero at infinity. Work done by gravity in bringing mass m from infinity to R is negative, giving $U = -GMm/R$.

4.5 State and explain work–energy theorem in a resistive medium.

The net work equals change in kinetic energy. With resistance, $mgh = \frac{1}{2}mv^2 + fh$, where fh is work done against resistive force; some mechanical energy becomes heat.

4.6 Define escape velocity. Show $v_e = \sqrt{(2Rg)}$ and find its value near Earth.

Escape velocity is the minimum speed to escape without further propulsion. $\frac{1}{2}mv_e^2 = GMm/R$ gives $v_e = \sqrt{(2GM/R)} = \sqrt{(2gR)}$. For Earth, $v_e \approx 11.2$ km s^{-1}.

Numerical Problems (fully worked)

4.1 A machine gun fires 6 bullets per minute at 700 m s^{-1}; each bullet mass 40 g. Find power.

Given: rate = 6/min = 0.1 s^{-1}, m = 0.04 kg, v = 700 m s^{-1}

Formula: $P = \text{rate} \times \frac{1}{2}mv^2$

$P = 0.1 \times \frac{1}{2} \times 0.04 \times 700^2 = 980$ W.

Answer: 980 W

4.2 A family uses 10 kW. Solar energy 300 W m^{-2}, 75% converted. Find area needed.

Given: P = 10,000 W, useful intensity = 0.75×300 = 225 W m^{-2}

Formula: A = P/intensity

$A = 10000/225 \approx 44.44$ m^2.

Answer: 44.44 m^2

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4.3 Earth mass 6.0×10^{24} kg, Sun mass 1.99×10^{30} kg, distance 160 million km. Find gravitational P.E. of Earth.

Given: $M = 1.99\times10^{30}$ kg, $m = 6.0\times10^{24}$ kg, $r = 1.6\times10^{11}$ m

Formula: $U = -GMm/r$

$U = -6.67\times10^{-11} \times 1.99\times10^{30} \times 6.0\times10^{24} / 1.6\times10^{11} \approx -4.97 \times 10^{33}$ J.

Answer: -4.97×10^{33} J

4.4 An object weighing 98 N dropped from 10 m hits ground at 12 m s^{-1}. Find frictional force.

Given: W = 98 N, m = 10 kg, h = 10 m, v = 12 m s^{-1}

Formula: $mgh = \frac{1}{2}mv^2 + fh$

$980 = 720 + f\times10$, so f = 26 N.

Answer: 26 N

4.5 A 75 W fan is used 8 h daily for 30 days. Find (i) energy in units (ii) bill at Rs. 22.5 per unit.

Given: P = 0.075 kW, $t = 8\times30 = 240$ h

Formula: $E = Pt$

$E = 0.075 \times 240 = 18$ kWh.

Bill = 18×22.5 = Rs. 405.

Answer: (i) 18 units (ii) Rs. 405

4.6 A 2 kg object is thrown up to 5 m and falls back. Find work by gravity (i) up (ii) down (iii) total.

Given: m = 2 kg, h = 5 m

Formula: W = −mgh upward, +mgh downward

(i) $-2\times9.8\times5 = -98$ J. (ii) +98 J. (iii) 0. Work in a closed path in a conservative field is zero.

Answer: (i) −98 J (ii) +98 J (iii) 0

4.7 A 1 hp motor runs a pump filling 400 L tank at 10 m in 15 min. Find (a) input work (b) output work.

Given: P = 746 W, t = 900 s, m = 400 kg, h = 10 m

Formula: $W_{in} = Pt$, $W_{out} = mgh$

$W_{in} = 746 \times 900 = 671.4$ kJ.

$W_{out} = 400 \times 9.8 \times 10 = 39.2$ kJ.

Answer: (a) 671.4 kJ (b) 39.2 kJ

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4.8 A passenger pulls a suitcase with 200 N at 45° to floor for 50 m. Find work done.

Given: F = 200 N, θ = 45°, d = 50 m

Formula: W = Fd cos θ

W = 200 × 50 × 0.707 ≈ 7070 J ≈ 7 kJ.

Answer: 7 kJ

4.9 A 1200 kg car at 40 km h^{-1} accelerates at 2 m s^{-2}. Find power expended.

Given: m = 1200 kg, v = 11.11 m s^{-1}, a = 2 m s^{-2}

Formula: P = Fv = mav

P = 1200 × 2 × 11.11 ≈ 26,667 W = 26.67 kW.

Answer: 26.67 kW

4.10 A 200 g apple is lifted 10 m and dropped. 75% of lifting work becomes K.E. Find impact velocity.

Given: m = 0.2 kg, h = 10 m, efficiency = 75%

Formula: ½mv^2 = 0.75mgh

v = √(2×0.75×9.8×10) = √147 ≈ 12.1 m s^{-1}.

Answer: 12.1 m s^{-1}

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